Yinda Xu

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Helsinki, Finland Mar. 1999, Ningbo, China

EDUCATION

Aalto University, Finland (QS 114) *Sept. 2023 – Present*
Doctoral Researcher, School of Electrical Engineering

- Smart Building Technologies and Services Group.
- Research focus: indoor sensing, ventilation–grid interaction, intelligent built environment.

Aalto University, Finland (QS 114) *Sept. 2021 – May 2023*
M.Sc. (Tech.), Automation and Electrical Engineering

- Major: Control, Robotics and Autonomous Systems. Department of Electrical Engineering and Automation.
- GPA: **4.29 / 5**. Graduated with Honours. **Full A-class scholarship** (top scholarship for international students).

Northeastern University, China (Project 985) *Sept. 2017 – June 2021*
B.Eng., School of Control Engineering

- Major: Automation. GPA: **3.7 / 5**. Class Rank: **23 / 130**.

INTERNSHIPS & WORK EXPERIENCE

Xiaomi Technology Inc. *Mar. – June 2021*
Android Development Engineer (Intern)

- Implemented new factory-mode features in the Mi TV Android system, supporting product-line testing and configuration.

Ningbo Meefo Software Co., Ltd. *Jan. – Feb. 2021*
Android Development Engineer (Intern)

- Designed a remote-trial Android app for local judicial agencies, focusing on document-camera integration and digital evidence capture.

Ningbo Boden AI Technology Co., Ltd. *July – Aug. 2020*
SLAM Development Engineer (Intern)

- Performed image processing with a ZED stereo camera and OpenCV (image-to-world coordinate transformation, pixel matching).
- Accelerated YOLOv4 inference for image recognition using NVIDIA TensorRT.

PROJECT EXPERIENCE

Occluded Indoor Target Tracking via Scattered mmWave Signals *Doctoral research*

- Tackled human–robot collision risk for mobile robots in NLoS scenarios (factories, warehouses, hospitals) by exploiting reflected and diffracted mmWave signals for blind-spot localization.
- Designed a Localization CNN (L-CNN) consuming 2D-FFT angle–delay heatmaps from FMCW radar to infer occluded-target position; applied a Kalman filter to improve tracking accuracy and robustness.
- Built a ground-truth annotation pipeline using an RGB-D camera and YOLO; achieved ∼**10 cm accuracy** without dedicated infrastructure.
- **Output:** *Tracking the Occluded Indoor Target with Scattered Millimeter Wave Signal*, IEEE Sensors Journal (first author, accepted).

Ventilation-based Primary Frequency Control for Power Grids *Doctoral research*

- Investigated using building HVAC systems as demand-side resources for grid Primary Frequency Control (PFC), motivated by reduced synchronous inertia from large-scale renewable integration.
- Proposed an open-loop AHU-based control strategy that leverages widespread variable-frequency drives (VFDs) and the large thermal inertia of indoor spaces.
- Validated in a controlled climate chamber; monitored CO_2, PM2.5 and TVOC, and ran standardized cognitive tests on 20 participants.
- Demonstrated good time- and frequency-domain dynamic response with no significant adverse impact on indoor air quality or cognitive performance.
- **Output:** *Investigating the Impact of Ventilation Primary Frequency Control on Indoor Climate and Cognitive Performance in Office Settings*, Building and Environment (first author, accepted).

Occupant Count Estimation via Deep Learning & CO_2 Sensors for Demand-Driven Ventilation *Doctoral research*

- Estimated occupant counts in a mechanically ventilated office room using a Convolutional Deep LSTM (CDLSTM) model

with CO_2 and airflow measurements; applied transfer learning with synthetic data to reduce training-data requirements.
- Achieved **79 % accuracy** for occupant counting; deployed the model online for occupancy-based demand-driven ventilation (DCV) with only marginal degradation (**75 %**), demonstrating feasibility under closed-loop control.
- Showed that random-setpoint ventilation in training data and shifting ground-truth labels (to compensate CO_2 propagation delay) jointly improve robustness, providing a pathway to lower HVAC energy consumption while maintaining indoor air quality.
- **Output:** J. Mikala, **Y. Xu**, M. Huotari, *Estimating occupant counts using deep neural networks and CO_2 sensors in a mechanically ventilated environment*, Energy and Buildings (second author; under review).

Building Knowledge Graph QA with Small Local LLMs (BuildingQA) *Doctoral research*

- Addressed the gap in the BuildingQA benchmark (BuildSys 2025), which only evaluates cloud-scale LLMs, by building an end-to-end SPARQL generation–execution–evaluation pipeline for small on-device models; covered 188 questions across 4 building knowledge graphs (Brick & ASHRAE 223P).
- Proposed a **schema-aware retrieval + arity-validation repair** pipeline and systematically compared LLM critique with LoRA fine-tuning; achieved **row_f1 = 0.321**, a big improvement over the strongest open-source baseline in the paper.

CURE-Bench Medical AI Competition *LLM agent for clinical decision support*

- Built an AI agent for drug decision-making and treatment planning using a multi-step reasoning workflow with tool calling and Retrieval-Augmented Generation (RAG).
- Integrated chain-of-thought prompting to improve LLM reasoning accuracy on medical knowledge tasks.

Droplet Detection and Tracking *Aalto University course project*

- Implemented and compared classical computer-vision pipelines (Canny edge detector, blob tracker) with deep-learning detectors (YOLO + DeepSORT) for droplet tracking under microscopy.

Mathematical Contest in Modeling (MCM/CUMCM) *Provincial & international*

- Modelled the pressure dynamics of an oil-tube system and derived an optimal inlet/outlet valve control strategy.
- Designed sandcastle structures in NX (UG), simulated tide-impact behaviour in ANSYS, and selected the most tide-resistant geometry.
- **Output: Meritorious Winner (M Award)**, 2020 Mathematical Contest in Modeling (MCM).

Intelligent Transportation Simulation & Control Lab *Northeastern University*

- Developed Arduino + OpenMV control software for autonomous cars: magnetic-line path following, traffic-light recognition, AprilTag-based safe inter-vehicle distance, and real-time motion control.
- Demo: A traffic dispatching simulation platform (Bilibili).

PUBLICATIONS

- **Y. Xu** et al., "Tracking the Occluded Indoor Target with Scattered Millimeter Wave Signal," *IEEE Sensors Journal* (accepted; first author; CAS Tier 3).
- **Y. Xu** et al., "Investigating the Impact of Ventilation Primary Frequency Control on Indoor Climate and Cognitive Performance in Office Settings," *Building and Environment* (accepted; first author; CAS Tier 1, Top).
- J. Nurmi, **Y. Xu**, J. Boutellier, B. Tan, "SPHERE-DNA: Privacy-Preserving Federated Learning for eHealth," in *Proc. Design, Automation & Test in Europe Conf. & Exhibition (DATE)*, Apr. 2023 (accepted; second author).
- J. Mikala, **Y. Xu**, M. Huotari, "Estimating Occupant Counts Using Deep Neural Networks and CO_2 Sensors in a Mechanically Ventilated Environment," *Energy and Buildings* (under review; second author; CAS Tier 2).
- **Y. Xu** et al., "A Systematic Review of Sensor-Based Machine Learning Methods for Building Occupancy Estimation," *IEEE Access* (under review; first author; CAS Tier 4).

HONORS & AWARDS

- Aalto University **A-class Full Scholarship** for Master's study (top tier for international students).
- **Meritorious Winner** (M Award), Mathematical Contest in Modeling (MCM), 2020.
- Undergraduate Entrepreneurship Scholarship – 2 times.
- Undergraduate Comprehensive Scholarship – Second class × 2, Third class × 3.

TECHNICAL SKILLS

Programming	C, C++, Java, Python, MATLAB.
Frameworks	PyTorch, TensorFlow, OpenCV, ROS, Git, Linux, TensorRT.
Domains	Embedded software/hardware, single-board hardware design, signal processing, machine learning, computer vision, robotics, reinforcement learning.
Languages	Chinese (native), English (professional), Finnish (basic).